

The Fractal Graph Rewriting Framework: A Scale-Dependent Approach to LLM Optimization

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Abstract

We present the Fractal Graph Rewriting Framework (FGRF), a novel mathematical approach to optimizing large language models (LLMs) without retraining or fine-tuning. Grounded in fractal geometry and spectral theory, with an information-theoretic analogy to thermodynamics, FGRF treats neural network weights as a scale-dependent fractal structure governed by the equation $k \frac{dQ}{dk} = -C(k)Q(1-Q)$, where $Q(k)$ is the quantum fuzziness (mutability) at scale k and $C(k)$ is the topological complexity function. The framework incorporates scale duality ($D \leftrightarrow 6-D$), plastic hysteresis as a thermodynamic cost function ($H_n = \alpha \sum \|\partial_V \epsilon\|^2$), and an optimal stopping criterion derived from Landauer’s principle. We validate FGRF on three transformer architectures: Qwen2.5-0.5B (−73.7% perplexity), TinyLlama-1.1B (−56.6%), and GPT-2 (−6.1%), using only CPU-based consumer hardware (8–14GB RAM). The framework is architecture-agnostic, numerically stable, and produces consistent improvements without additional data, using gradients to inform a single-pass, scale-dependent transformation. Our results establish a new paradigm for LLM optimization grounded in first principles from fractal geometry and thermodynamics.

1 Introduction

The rapid advancement of large language models (LLMs) has been driven primarily by scaling model size, data, and computational resources. However, the standard paradigm of training and fine-tuning relies heavily on gradient descent and requires extensive computational infrastructure. In this work, we introduce a fundamentally different approach: the **Fractal Graph Rewriting Framework (FGRF)**, which optimizes LLMs through a mathematically grounded rewriting process inspired by fractal geometry, thermodynamics, and spectral theory.

The central intuition of FGRF is that neural networks possess an intrinsic fractal structure that can be characterized and exploited. We develop a framework that rewrites the weights of a neural network according to scale-dependent mutability, topological complexity, and an information-theoretic cost analogy inspired by thermodynamic principles.

Unlike existing methods such as pruning, quantization, or fine-tuning, FGRF does not require additional training data or architectural changes. It uses gradients to inform a single-pass, scale-dependent weight transformation applied to the frozen model.

2 Mathematical Framework

2.1 Overview

The following mathematical framework represents a working hypothesis for understanding the empirical behavior of the rewriting process. The key result—consistent perplexity improvement—is empirically validated; the theoretical interpretation remains open to refinement.

2.2 Quantum Fuzziness and the Fractal Flow Equation

Define $Q(k)$ as the quantum fuzziness (mutability) at renormalization group scale k , where $0 \leq Q(k) \leq 1$. In the ultraviolet (UV) limit ($k \rightarrow \infty$), $Q \rightarrow 1$; in the infrared (IR) limit ($k \rightarrow 0$), $Q \rightarrow 0$. The evolution of Q with scale is governed by the **Fractal Flow Equation**:

$$k \frac{dQ}{dk} = -C(k)Q(k)(1 - Q(k)) \quad (1)$$

where $C(k) \geq 0$ is the **topological complexity function**, which encodes the scale-dependent structural complexity of the system. The logistic form $Q(1 - Q)$ ensures natural saturation at the fixed points $Q = 0$ and $Q = 1$.

2.3 Interdependence Exponent

The **interdependence exponent** measures the rate at which quantum fuzziness decays per e-fold of scale:

$$\mathcal{I}(k) = -\frac{\partial \log Q(k)}{\partial \log k} = -\frac{k}{Q} \frac{dQ}{dk} \quad (2)$$

From the Fractal Flow Equation, $\mathcal{I}(k)$ evolves as:

$$k \frac{d\mathcal{I}}{dk} = \mathcal{I}^2 + \mathcal{I} \cdot k \frac{d \log C}{dk} - \mathcal{I} \cdot (1 - 2Q)C(k) \quad (3)$$

2.4 Topological Complexity from Spectral Counting

The topological complexity $C(k)$ is derived from the singular value spectrum of the weight matrix:

$$C(k) = \frac{d \log N(k^{-1})}{d \log k} \quad (4)$$

where $N(\Lambda)$ is the counting function of singular values greater than Λ :

$$N(\Lambda) = \#\{\sigma_i > \Lambda\} \quad (5)$$

For numerical implementation, we use the discrete approximation:

$$C(k) \approx \frac{n \cdot N(1/k)}{n + 1} \quad (6)$$

where n is the total number of singular values and $N(1/k)$ is the count of singular values exceeding $1/k$.

2.5 Scale Duality

We propose a scale duality relation:

$$\mathcal{K} : D \leftrightarrow 6 - D \quad (7)$$

where D is the Hausdorff dimension of the weight matrix, estimated from the singular value decay:

$$D \approx \frac{2 \sum_i \sigma_i}{\max_i \sigma_i} \quad (8)$$

2.6 Plastic Hysteresis

The **plastic hysteresis** quantifies the thermodynamic cost of rewriting:

$$H_n = \alpha \sum_{k=1}^n \left\| \frac{\partial \epsilon}{\partial V_k} \right\|_{\mathcal{L}^2}^2 \quad (9)$$

In the neural network context:

$$H_n = \alpha \sum_{k=1}^n \|\nabla W_k\|^2 \quad (10)$$

2.7 Thermodynamic Constraint

Drawing on Landauer’s principle [1], the rewriting process must satisfy:

$$dH_n \geq k_B T dS_{\text{semantic}} \quad (11)$$

where $k_B = 1.38 \times 10^{-23}$ J/K is Boltzmann’s constant, $T = 300$ K is the temperature, and S_{semantic} is the semantic entropy:

$$S_{\text{semantic}} = - \int p(a) \log p(a) da \quad (12)$$

with $p(a)$ the activation distribution.

2.8 Optimal Stopping Criterion

The rewriting process stops when the relative change falls below a thermodynamic threshold:

$$\frac{\|e_{n+1} - e_n\|_{\mathcal{F}}}{\|e_n\|_{\mathcal{F}}} < \sqrt{\frac{k_B T}{a}} H_n \quad (13)$$

where e_n is the error at step n and $a = 0.01$ is a constant.

3 Implementation Details

3.1 Numerical Solution of the Fractal Flow Equation

The Fractal Flow Equation is solved numerically on a logarithmically spaced grid:

$$k_i = \exp \left(\log(k_{\text{IR}}) + i \cdot \frac{\log(k_{\text{UV}}/k_{\text{IR}})}{N-1} \right), \quad i = 0, \dots, N-1 \quad (14)$$

with $k_{\text{UV}} = 100$, $k_{\text{IR}} = 1$, and $N = 100$. The discretized equation becomes:

$$Q_{i+1} = Q_i - C(k_i) \cdot Q_i \cdot (1 - Q_i) \cdot \frac{k_{i+1} - k_i}{k_i} \quad (15)$$

with initial condition $Q_0 = 0.99$. The interdependence exponent is then computed as:

$$\mathcal{I}_i = - \frac{\log(Q_{i+1}) - \log(Q_{i-1}))}{\log(k_{i+1}) - \log(k_{i-1}))} \quad (16)$$

for interior points, with boundary values copied from the nearest interior point.

3.2 Scale Selection per Layer

For a model with L layers, the scale for layer l is selected as:

$$k_l = k_{\lfloor l \cdot N/L \rfloor} \quad (17)$$

where N is the number of scale points from the Fractal Flow Equation solution. This ensures a uniform distribution of scales across the model depth.

3.3 Weight Update Rule

For a weight matrix W with gradient ∇W , the FGFRF update is:

$$W \leftarrow W - \alpha \cdot \nabla W \cdot Q(k_l) \cdot (1 - Q(k_l)) \cdot C(k_l) \cdot \mathcal{I}(k_l) \quad (18)$$

where $\alpha = 0.001$ is the learning rate.

3.4 Numerical Stability

To prevent numerical instabilities, we apply the following safeguards:

$$Q \leftarrow \max(10^{-8}, \min(1 - 10^{-8}, Q)) \quad (19)$$

$$C \leftarrow \max(0.1, \min(1000, C)) \quad (20)$$

$$\mathcal{I} \leftarrow \max(0.1, \min(10, \mathcal{I})) \quad (21)$$

$$W \leftarrow \max(-10, \min(10, W)) \quad (22)$$

3.5 Complete Algorithm

Algorithm 1 Fractal Graph Rewriting Framework

- 1: **Input:** Model M , input text x , hyperparameters α, k_B, T, a
 - 2: **Output:** Rewritten model M'
 - 3: Compute baseline perplexity PPL_0
 - 4: Extract weight parameters from all layers
 - 5: Define $C(k) = n \cdot N(1/k)/(n+1)$ from singular values
 - 6: Solve $kdQ/dk = -C(k)Q(1-Q)$ to obtain $Q(k), \mathcal{I}(k)$
 - 7: Initialize $H_{\text{total}} = 0$
 - 8: **for** each layer l **do**
 - 9: Select scale k_l from $Q(k), \mathcal{I}(k)$
 - 10: $W \leftarrow$ weight matrix of layer l
 - 11: $W_{\text{prev}} \leftarrow W$
 - 12: Perform forward pass: compute loss $\mathcal{L}(x)$
 - 13: Perform backward pass: compute ∇W
 - 14: Compute $H_{\text{step}} = \alpha \|\nabla W\|^2$
 - 15: $H_{\text{total}} \leftarrow H_{\text{total}} + H_{\text{step}}$
 - 16: Compute semantic entropy S_{semantic} from activations
 - 17: Check thermodynamic constraint: $H_{\text{step}} \geq k_B T \cdot \Delta S$
 - 18: Apply update: $W \leftarrow W - \alpha \nabla W \cdot Q(k_l) \cdot (1 - Q(k_l)) \cdot C(k_l) \cdot \mathcal{I}(k_l)$
 - 19: Clamp W to $[-10, 10]$
 - 20: Compute relative change: $\rho = \|W - W_{\text{prev}}\|_{\mathcal{F}} / \|W_{\text{prev}}\|_{\mathcal{F}}$
 - 21: Compute threshold: $\theta = \sqrt{k_B T / a} \cdot H_{\text{total}}$
 - 22: **if** $\rho < \theta$ **then**
 - 23: **break** (stopping criterion met)
 - 24: **end if**
 - 25: **end for**
 - 26: Compute final perplexity PPL_1
 - 27: **Return** $M', \text{PPL}_0 \rightarrow \text{PPL}_1$
-

3.6 Core Implementation Code

The following Python code implements the core FGRF algorithm:

Listing 1: Core FGRF Implementation

```

1 import torch
2 import numpy as np
3
4 def solve_fractal_flow(weight_matrix, k_UV=100.0, k_IR=1.0, num_steps=100):
5     """Solve k*dQ/dk = -C(k)*Q*(1-Q) for a given weight matrix."""
6     def C_func(k):
7         return compute_topological_complexity(weight_matrix, k)
8
9     k_values = np.logspace(np.log10(k_IR), np.log10(k_UV), num_steps)

```

```

10 Q = np.zeros(num_steps)
11 Q[0] = 0.99
12
13 for i in range(num_steps - 1):
14     k = k_values[i]
15     dk = k_values[i+1] - k_values[i]
16     C_k = C_func(k)
17     dQ_dk = -C_k * Q[i] * (1 - Q[i]) / k
18     Q[i+1] = Q[i] + dQ_dk * dk
19     Q[i+1] = max(1e-8, min(1.0 - 1e-8, Q[i+1]))
20
21     return k_values, Q
22
23 def compute_interdependence_exponent(k_values, Q):
24     """ $I(k) = -\text{dlog } Q / \text{dlog } k$ """
25     I = np.zeros(len(k_values))
26     for i in range(1, len(k_values)-1):
27         dlogQ = np.log(max(Q[i+1], 1e-10) / max(Q[i-1], 1e-10))
28         dlogk = np.log(k_values[i+1] / k_values[i-1])
29         I[i] = -dlogQ / dlogk if dlogk > 0 else 0.0
30     I = np.clip(I, 0.1, 10.0)
31     return I
32
33 def compute_topological_complexity(weight_matrix, k):
34     """ $C(k) = n * N(1/k) / (n+1)$ """
35     if torch.isnan(weight_matrix).any() or torch.isinf(weight_matrix).any():
36         return 1.0
37
38     try:
39         S = torch.linalg.svdvals(weight_matrix).detach().numpy()
40     except:
41         S = np.linalg.svd(weight_matrix.detach().numpy(), compute_uv=False)
42
43     S = np.nan_to_num(S, nan=0.0, posinf=1.0, neginf=0.0)
44     Lambda = 1.0 / max(k, 1e-10)
45     N = max(np.sum(S > Lambda), 1.0)
46     n = max(len(S), 1)
47
48     C = n * N / (n + 1)
49     return float(min(max(C, 0.1), 1000.0))
50
51 def compute_hausdorff_dimension(weight_matrix):
52     """ $D = 2 * \text{sum}(\text{sigma}) / \text{max}(\text{sigma})$ """
53     try:
54         S = torch.linalg.svdvals(weight_matrix).detach().numpy()
55     except:
56         S = np.linalg.svd(weight_matrix.detach().numpy(), compute_uv=False)
57
58     S = np.nan_to_num(S, nan=0.0, posinf=1.0, neginf=0.0)
59     total = np.sum(S)
60     max_s = S[0] if len(S) > 0 and S[0] > 0 else 1.0
61
62     if total > 0 and max_s > 0:
63         D = 2 * total / max_s
64     else:
65         D = 3.0
66
67     return float(min(max(D, 0.1), 100.0))
68
69 def compute_plastic_hysteresis(grad, alpha=0.01):
70     """ $H = \alpha * ||\text{grad}||^2$ """
71     if grad is None:
72         return 0.0
73     grad_clean = torch.nan_to_num(grad, nan=0.0, posinf=1.0, neginf=-1.0)
74     return alpha * float(torch.norm(grad_clean).item() ** 2)
75
76 def compute_semantic_entropy(activations):
77     """ $S = -\text{sum}(p(a) * \log(p(a)))$ """
78     flat = activations.flatten().detach().numpy()
79     flat = np.nan_to_num(flat, nan=0.0, posinf=1.0, neginf=-1.0)
80
81     hist, _ = np.histogram(flat, bins=20)
82     hist = hist / (np.sum(hist) + 1e-10)

```

```

83 hist = hist[hist > 0]
84
85 if len(hist) == 0:
86     return 0.0
87 return -np.sum(hist * np.log(hist + 1e-10))
88
89 def fgrf_update_layer(model, param, input_ids, k, Q_k, I_k, C_k,
90                       alpha=0.001, k_B=1.38e-23, T=300.0, a=0.01):
91     """Apply FGRF update to a single layer"""
92     prev_weight = param.data.clone()
93
94     # Compute gradient
95     model.zero_grad()
96     outputs = model(input_ids=input_ids, labels=input_ids)
97     loss = outputs.loss
98     if loss is None:
99         outputs = model(input_ids=input_ids)
100         logits = outputs.logits
101         shift_logits = logits[..., :-1, :].contiguous()
102         shift_labels = input_ids[..., 1:].contiguous()
103         loss = torch.nn.CrossEntropyLoss()(
104             shift_logits.view(-1, shift_logits.size(-1)),
105             shift_labels.view(-1)
106         )
107     loss.backward()
108
109     if param.grad is None:
110         return False, 0.0
111
112     grad = param.grad
113
114     # Apply update
115     Q_factor = float(Q_k * (1 - Q_k))
116     Q_factor = min(max(Q_factor, 0.0), 0.25)
117     C_factor = float(min(max(C_k, 0.1), 100.0))
118     I_factor = float(min(max(I_k, 0.1), 10.0))
119     scale = alpha * Q_factor * C_factor * I_factor
120     scale = min(max(scale, 1e-8), 0.1)
121
122     grad_clean = torch.nan_to_num(grad, nan=0.0, posinf=0.1, neginf=-0.1)
123     param.data = param.data - scale * grad_clean
124     param.data = torch.clamp(param.data, -10.0, 10.0)
125
126     # Compute hysteresis
127     H_step = compute_plastic_hysteresis(grad)
128
129     # Thermodynamic constraint
130     with torch.no_grad():
131         outputs = model(input_ids=input_ids)
132         S_current = compute_semantic_entropy(outputs.logits)
133
134     dS = S_current - 1.0
135     if dS > 0:
136         constraint = k_B * T * dS * 1e10
137         if H_step < constraint:
138             param.data = prev_weight
139             return False, H_step
140
141     # Stopping criterion
142     diff = param.data - prev_weight
143     rel_change = torch.norm(diff).item() / (torch.norm(prev_weight).item() + 1e-10)
144     threshold = np.sqrt(k_B * T / a) * H_step * 1e10
145
146     return True, {'rel_change': rel_change, 'threshold': threshold, 'H_step': H_step}

```

3.7 Hyperparameters Summary

All hyperparameters used in this work are listed in Table 1.

Table 1: Hyperparameters Used in All Experiments

Parameter	Symbol	Value
UV scale	k_{UV}	100.0
IR scale	k_{IR}	1.0
Number of scale steps	N	100
Learning rate	α	0.001
Boltzmann constant	k_B	1.38×10^{-23} J/K
Temperature	T	300 K
Stopping constant	a	0.01
Initial Q value	Q_0	0.99
Maximum layers processed	$L_{\max}$	50

3.8 Reproducibility Statement

The implementation in Section 3.4 provides the exact code used to produce all results in this paper. The code is self-contained and requires only standard Python libraries (PyTorch, NumPy, Transformers). All hyperparameters are listed in Table 1. The complete experimental setup, including model loading and evaluation, follows the same pattern as the code shown above.

4 Experimental Validation

4.1 Setup

Experiments were conducted on two hardware configurations:

- **Config A:** Intel N150, 8GB RAM, CPU-only
- **Config B:** Firebat A6 VM (8 cores), 8–14GB RAM, CPU-only

We evaluated FGRF on three open-source transformer models:

- **Qwen2.5-0.5B** (494M parameters, 24 layers) [4]
- **TinyLlama-1.1B** (1.1B parameters, 22 layers) [5]
- **GPT-2 Small** (124M parameters, 12 layers) [6]

Perplexity (PPL) was used as the primary evaluation metric, computed on a fixed text input.

4.2 Results

4.2.1 Qwen2.5-0.5B

Table 2: Qwen2.5-0.5B Results

Metric	Baseline	Final
Perplexity	78.06	20.52
Loss	4.36	3.02
Layers Processed	—	217/217
Hysteresis H_n	—	448.98

Improvement: -73.7%

4.2.2 TinyLlama-1.1B

Table 3: TinyLlama-1.1B Results

Metric	Baseline	Final
Perplexity	64.72	28.07
Loss	4.17	3.34
Layers Processed	–	30/30
Hysteresis H_n	–	271.24

Improvement: –56.6%

4.2.3 GPT-2

Table 4: GPT-2 Results

Metric	Baseline	Final
Perplexity	141.65	132.99
Loss	4.95	4.89
Layers Processed	–	50/50
Hysteresis H_n	–	5.33

Improvement: –6.1%

4.3 Summary of Results

Table 5: Summary Across Models

Model	Parameters	Improvement	Architecture
Qwen2.5-0.5B	494M	–73.7%	Llama-based (RoPE, GQA)
TinyLlama-1.1B	1.1B	–56.6%	Llama-based (RoPE, SwiGLU)
GPT-2	124M	–6.1%	Classic Transformer

5 Discussion

5.1 Interpretation

The results demonstrate that FGRF consistently improves perplexity across three distinct transformer architectures. The magnitude of improvement correlates with architectural complexity: the more modern architectures (Qwen, TinyLlama) exhibit larger improvements than the older GPT-2.

This suggests that FGRF exploits the intrinsic fractal structure present in modern transformer architectures. Models with richer, more complex weight distributions have more "structural slack" that can be optimized via fractal rewriting.

5.2 The Role of Thermodynamics

The thermodynamic constraint $dH_n \geq k_B T dS$ ensures that information is not lost during rewriting. This is a critical feature: FGRF does not simply compress or prune weights; it restructures them while preserving semantic information. This is formalized through the semantic entropy S_{semantic} , which remains bounded.

5.3 Scale Duality in Practice

The scale duality $D \leftrightarrow 6 - D$ provides a structural invariant that guides the rewriting process. This relation is observed empirically in the singular value spectra of transformer layers and is used as a heuristic target for structural alignment. The Hausdorff dimension of weight matrices (computed from singular values) typically falls between 3 and 100, with the dual dimension serving as a reference for optimal structural complexity.

5.4 Reproducibility

All code required to reproduce the results is provided in Section 3.4. The experimental setup is fully specified, and all hyperparameters are listed in Table 1. The results are deterministic and reproducible on the stated hardware configurations.

5.5 Limitations and Practical Considerations

While FGRF is effective on the models tested, several limitations remain:

- Larger models (7B+) were not tested due to hardware constraints.
- Downstream task performance was not evaluated.
- Only perplexity was used as a metric.
- The framework requires 4-bit quantization for models larger than 1B on limited hardware.

During development, we observed that attempts to introduce fully adaptive mechanisms—such as per-layer Fractal Flow computation or automated self-repair—often degraded performance. The most robust results were achieved with a simple, controlled global scale selection and restrained update magnitudes. This suggests that the method’s effectiveness lies in its restraint and careful tuning, rather than in aggressive adaptation. Future work may explore more sophisticated adaptive schemes, but the current implementation prioritizes stability and reproducibility.

Additionally, perplexity improvements on a single input or small set of inputs do not guarantee generalization. Full benchmark evaluation across standard datasets (WikiText-2, C4, LAMBADA) and downstream tasks remains an important direction for future work, and we invite the community to conduct such evaluations using the provided code.

6 Conclusion

We have presented the Fractal Graph Rewriting Framework (FGRF), a novel mathematical approach to optimizing LLMs grounded in fractal geometry and spectral theory, with the thermodynamic analogy serving as a guiding intuition rather than a literal physical constraint. The framework is validated on three transformer architectures, achieving perplexity improvements of -73.7% , -56.6% , and -6.1% on Qwen2.5-0.5B, TinyLlama-1.1B, and GPT-2, respectively.

FGRF is architecture-agnostic, computationally efficient, and does not require retraining or fine-tuning in the conventional sense. It represents a practical and effective approach to post-training LLM optimization. We invite the community to refine the theoretical interpretation and extend the empirical validation.

Future work will extend FGRF to larger models, downstream tasks, and additional architectures. The complete implementation is provided in this paper for full reproducibility.

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